

ESC201: Mechanics of Materials

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Take Home Exam

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1. Question 1

1.a. Given Values and Model

The spar is modeled as a cantilever beam with a deformable plate/spar joint. Define the origin of the beam ($x=0$) to be the point where the spar is fastened to the wall. The center of the airframe is assumed rigid, so the boundary conditions of the spar are:

$$y(0) = 0, \quad \theta(0) = 0$$

The maximum thrust is

$$F_T = 80 \text{ N}$$

The relevant dimensions and material properties are

$$t_p = 4 \text{ mm}, \quad w_p = 44 \text{ mm}, \quad t_w = 2 \text{ mm}, \quad d_o = 25 \text{ mm}$$

$$L_b = 76 \text{ mm}, \quad L_s = 400 \text{ mm}, \quad E_{CF,0} = 40 \text{ GPa}, \quad \sigma_{u,0} = 800 \text{ MPa}$$

The total spar length from the rigid airframe center to the motor is:

$$L = L_b + L_s$$

For this simplified design, the spar and plates are both treated as $0^\circ/90^\circ$ carbon fiber, so:

$$E = E_{CF,0} = 40 \text{ GPa}$$

1.b. Section Properties

The spar is modeled as a square hollow tube. The inner side length is

$$d_i = d_o - 2t_w$$

The spar moment of inertia is therefore:

$$I_s = \frac{d_o^4 - d_i^4}{12} = \frac{d_o^4 - (d_o - 2t_w)^4}{12}$$

Each plate has area:

$$A_p = w_p t_p$$

The centroidal moment of inertia of one plate about its own centroidal axis is:

$$I_{p,c} = \frac{w_p t_p^3}{12}$$

The distance from the neutral axis of the spar to the centroid of each plate is

$$d_p = \frac{1}{2}(d_o + t_p)$$

Using the parallel-axis theorem, the moment of inertia of the joint region is:

$$I_j = I_s + 2(I_{p,c} + A_p d_p^2)$$

Thus, the bending stiffnesses are:

$$EI_s = 653.8 \text{ N} \cdot \text{m}^2$$

$$EI_j = 3632.9 \text{ N} \cdot \text{m}^2$$

1.c. Derivation of Slope and Deflection

Consider the joint region, where $0 \leq x \leq L_b$. In this region, the bending moment is:

$$M(x) = F_T((L_b + L_s) - x)$$

Using the Euler-Bernoulli beam relation,

$$\frac{d^2 y}{dx^2} = \frac{M(x)}{EI_j} = \frac{F_T}{EI_j}((L_b + L_s) - x)$$

Integrating once gives the slope:

$$\theta(x) = \frac{F_T}{EI_j} \left((L_b + L_s)x - \frac{x^2}{2} \right) + C_1$$

Using the boundary condition on θ , we can find the value for C_1

$$0 = \theta(0) = \frac{F_T}{EI_j}(0) + C_1 = C_1$$

Therefore:

$$\theta(x) = \frac{F_T}{EI_j} \left((L_b + L_s)x - \frac{x^2}{2} \right)$$

At $x = L_b$:

$$\theta(L_b) = \frac{F_T}{EI_j} \left((L_b + L_s)L_b - \frac{L_b^2}{2} \right) = \frac{F_T}{EI_j} \left(L_s L_b + \frac{L_b^2}{2} \right) = \frac{F_T L_b}{EI_j} \left(L_s + \frac{L_b}{2} \right)$$

Integrating $\theta(x)$ second time gives the deflection:

$$y(x) = \frac{F_T}{EI_j} \left((L_b + L_s) \frac{x^2}{2} - \frac{x^3}{6} \right) + C_2$$

Using the boundary condition on y , we can find the value of C_2

$$0 = y(0) = \frac{F_T}{EI_j}(0) + C_2 = C_2$$

Thus,

$$y(x) = \frac{F_T}{EI_j} \left((L_b + L_s) \frac{x^2}{2} - \frac{x^3}{6} \right)$$

At $x = L_b$,

$$y(L_b) = \frac{F_T}{EI_j} \left((L_b + L_s) \frac{L_b^2}{2} - \frac{L_b^3}{6} \right) = \frac{F_T}{EI_j} \left(\frac{L_s L_b^2}{2} + \frac{L_b^3}{3} \right) = \frac{F_T L_b^2}{EI_j} \left(\frac{L_s}{2} + \frac{L_b}{3} \right)$$

For the free spar region, the additional slope and deflection from the free spar length are the standard cantilever end-load results (refer to Appendix D, “Beam Deflections and Shapes”, of the 4th edition of *Mechanics of Materials* by Beer et. al):

$$\theta(L_s) = \frac{F_T L_s^2}{2EI_s}$$

$$y(L_s) = \frac{F_T L_s^3}{3EI_s}$$

Therefore, the total tip slope is

$$\theta_{\text{tip}} = \theta(L_b) + \frac{F_T L_s^2}{2EI_s} = \frac{F_T L_b}{EI_j} \left(L_s + \frac{L_b}{2} \right) + \frac{F_T L_s^2}{2EI_s}$$

and the total tip deflection is

$$\begin{aligned} y_{\text{tip}} &= y(L_b) + \theta(L_b) L_s + \frac{F_T L_s^3}{3EI_s} \\ &= \frac{F_T L_b^2}{EI_j} \left(\frac{L_s}{2} + \frac{L_b}{3} \right) + \frac{F_T L_s L_b}{EI_j} \left(L_s + \frac{L_b}{2} \right) + \frac{F_T L_s^3}{3EI_s} \\ &= \frac{F_T L_s L_b}{EI_j} + \frac{F_T L_b^3}{3EI_j} + \frac{F_T L_s^2 L_b}{EI_j} + \frac{F_T L_s^3}{3EI_s} \end{aligned}$$

1.d. Numerical Results

At $x = L_b$:

$$\theta(L_b) = \frac{F_T L_b}{EI_j} \left(L_s + \frac{L_b}{2} \right) = 7.33 \times 10^{-4} \text{ rad} = 0.0420^\circ$$

The deflection at the end of the plate/spar joint is:

$$y(L_b) = \frac{F_T L_b^2}{EI_j} \left(\frac{L_s}{2} + \frac{L_b}{3} \right) = 2.866 \times 10^{-5} \text{ m} = 0.0287 \text{ mm}$$

At the motor tip:

$$\theta_{\text{tip}} = \theta(L_b) + \frac{F_T L_s^2}{2EI_s} = 1.052 \times 10^{-2} \text{ rad} = 0.603^\circ$$

Therefore, the total tip deflection is:

$$y_{\text{tip}} = \frac{F_T L_s L_b}{EI_j} + \frac{F_T L_b^3}{3EI_j} + \frac{F_T L_s^2 L_b}{EI_j} + \frac{F_T L_s^3}{3EI_s} = 2.932 \times 10^{-3} \text{ m} = 2.93 \text{ mm}$$

Location	Slope	Slope	Deflection
$x = L_b$	$7.33 \times 10^{-4} \text{ rad}$	0.0420°	0.0287 mm
$x = L_b + L_s$	$1.052 \times 10^{-2} \text{ rad}$	0.603°	2.93 mm

Table 1: Calculated slope and deflection at the end of the joint and at the motor tip.

1.e. Simple Checks

First, if the plate/spar joint were perfectly rigid, only the free spar length L_s would bend. This gives a lower-bound estimate:

$$y_{\text{lower}} = \frac{F_T L_s^3}{3EI_s} = 2.61 \text{ mm}$$

$$\theta_{\text{lower}} = \frac{F_T L_s^2}{2EI_s} = 9.79 \times 10^{-3} \text{ rad}$$

Second, if the entire length $(L_b + L_s)$ behaved like the spar alone, the deflection would be:

$$y_{\text{upper}} = \frac{F_T (L_b + L_s)^3}{3EI_s} = 4.40 \text{ mm}$$

$$\theta_{\text{upper}} = \frac{F_T(L_b + L_s)^2}{2EI_s} = 1.39 \times 10^{-2} \text{ rad}$$

The calculated result is

$$y_{\text{lower}} < 2.93 \text{ mm} < y_{\text{upper}}$$

$$\theta_{\text{lower}} < 1.052 \times 10^{-2} \text{ rad} < \theta_{\text{upper}}$$

This makes physical sense, as the real structure is more flexible than a perfectly rigid joint, but stiffer than a full-length spar-only cantilever.

1.f. Factor of Safety

Since the carbon fiber is brittle, I use a maximum normal stress failure criterion:

$$n := \frac{\sigma_u}{\sigma_{\text{max}}}$$

The maximum bending moment occurs at the fixed end:

$$M(0) = F_T(L_b + L_s) = 38.08 \text{ N} \cdot \text{m}$$

The outer fiber distance for the joint is:

$$c_j = \frac{d_o}{2} + t_p$$

Thus, the maximum normal stress in the joint region is:

$$\sigma_j = \frac{M(0)c_j}{I_j} = 6.92 \text{ MPa}$$

For the free spar region, the maximum bending moment occurs at $x = L_b$:

$$M(L_b) = F_T L_s = 32.0 \text{ N} \cdot \text{m}$$

The outer fiber distance for the spar is:

$$c_s = \frac{d_o}{2}$$

The bending stress in the free spar is:

$$\sigma_s = \frac{M(L_b)c_s}{I_s} = 24.5 \text{ MPa}$$

The free spar controls because:

$$\sigma_s > \sigma_j$$

Therefore,

$$n = \frac{\sigma_{u,0}}{\sigma_s}$$

$$n = \frac{800 \text{ MPa}}{24.5 \text{ MPa}} = 32.7$$

1.g. Discussion

The estimated maximum tip deflection is

$$y_{\text{tip}} \approx 2.93 \text{ mm}$$

This is small compared with the free spar length of 400 mm:

$$\frac{y_{\text{tip}}}{L_s} = \frac{2.93}{400} = 0.00733$$

or about 0.73% of the free spar length. The static deflection therefore seems reasonably small.

Recall that the factor of safety against carbon-fiber bending failure is:

$$n \approx 32.7$$

which is very large. Thus, the carbon fiber itself is not close to brittle bending failure under the maximum thrust load. However, the original motivation for this calculation is vibration, and a static deflection calculation does not by itself guarantee that the natural frequency is high enough. The more concerning real failure modes are probably joint slip, screw bearing, local crushing, delamination near the holes, fatigue, or low-frequency vibration rather than gross carbon-fiber bending failure.

1.h. Extra Credit: Rough Estimate for Original

Socket Design

For the original socket design, I estimate the joint stiffness by including the carbon-fiber spar, the 45° carbon-fiber plates, and the ASA socket material.

The plate modulus is:

$$E_{CF,45} = 10 \text{ GPa}$$

and the socket modulus is:

$$E_{\text{ASA}} = 2.1 \text{ GPa}$$

A rough square ASA collar model uses:

$$I_{\text{ASA}} = \frac{(d_o + 2t_{\text{ASA}})^4 - d_o^4}{12}$$

This gives a rough original joint flexural rigidity of:

$$EI_{\text{j, orig}} = E_{\text{CF},0}I_s + E_{\text{ASA}}I_{\text{ASA}} + 2E_{\text{CF},45} \left(I_{p,c} + A_p d_p^2 + \frac{A_p}{2} (d_0 + t_p)^2 \right) \approx 2191 \text{ N} \cdot \text{m}^2$$

Recall that the simplified new joint had

$$EI_{\text{j}} = 3632.9 \text{ N} \cdot \text{m}^2$$

Using the same beam formulas, we can

$$y_{\text{tip, orig}} = y(L_b) + \theta(L_b)L_s + \frac{F_T L_s^3}{3EI_s} \approx 3.14 \text{ mm}$$

The ratio is

$$\frac{y_{\text{tip, orig}}}{y_{\text{tip, new}}} = \frac{3.14}{2.93} = 1.07$$

Thus, this rough estimate predicts that the original socket design is about 7% more flexible than the simplified new design. This difference is not large because most of the deflection comes from the long free spar, not from the short joint region.

2. Question 2

2.a. Screw Shear Force

The simplified design uses 8 screws to hold the plates and spar together. The problem states to assume that the screws bear equal shear loads. Therefore, the shear load per screw is estimated by dividing the applied thrust force equally among the 8 screws:

$$V_{\text{screw}} = \frac{F_T}{8 \text{ screws}} = \frac{10 \text{ N}}{\text{screw}}$$

2.b. Shear Stress in One Screw

Each screw has diameter:

$$d = 3 \text{ mm}$$

The cross-sectional area of one screw is:

$$A_s = \frac{\pi d^2}{4}$$

The average shear stress in one screw is therefore:

$$\tau_{\text{screw}} = \frac{V_{\text{screw}}}{A_s} = \frac{10 \text{ N}}{7.07 \text{ mm}^2} = 1.41 \text{ MPa}$$

2.c. Discussion

A shear stress of only

$$\tau_{\text{screw}} \approx 1.41 \text{ MPa}$$

is very small for a typical 3 mm metal screw. Therefore, the screw shank itself is unlikely to fail in simple shear. The more concerning failure modes would be bearing failure in the carbon-fiber plate, local crushing around the screw holes, tear-out, delamination, or loosening of the bolted joint.

As a conservative composite-action check, one can also estimate the shear flow needed to make one plate act compositely with the spar:

$$q = \frac{VQ}{I_j}$$

where:

$$Q = A_p d_p = 2.552 \times 10^{-6} \text{ m}^3$$

Thus,

$$q = \frac{VQ}{I_j} = 2248 \text{ N/m}$$

Over the overlap length:

$$F_{\text{total}} = 2F_{\text{plate}} = 2qL_b = 341.6 \text{ N}$$

If this shear-transfer force were shared by 8 screws,

$$V_{\text{screw}} = \frac{341.6 \text{ N}}{8 \text{ screws}} = \frac{42.71 \text{ N}}{\text{screw}}$$

Even this more conservative estimate gives

$$\tau_{\text{screw,flow}} = \frac{42.71 \text{ N/screw}}{A_s} = 6.04 \text{ MPa}$$

which is still a small shear stress for a 3 mm metal screw. Therefore, either estimate suggests that direct screw shear is not the limiting issue.

2.d. Extra Credit: Original Socket Design

For the original socket design, the screw load can be estimated in the same way as the conservative shear-flow estimate above. The key difference is that the original design has only 4 screws instead of 8, and the plate/socket/spar joint has a different effective bending stiffness.

From the rough socket estimate in Question 1,

$$(EI)_{j,\text{orig}} \approx 2191 \text{ N} \cdot \text{m}^2$$

Because the original plate is separated from the spar by the ASA socket, the centroid of each plate is farther from the neutral axis. Using a socket thickness of 5 mm between each plate and the spar,

$$d_{p,\text{orig}} = \frac{d_o}{2} + t_{\text{ASA}} + \frac{t_p}{2} = 19.5 \text{ mm}$$

Since the original plate uses the 45° carbon-fiber orientation, use:

$$E_{CF,45} = 10 \text{ GPa}$$

For a multi-material composite section, the shear flow transferred into one plate can be estimated using the modulus-weighted form:

$$q_{\text{orig}} = \frac{V E_{CF,45} A_p d_{p,\text{orig}}}{E I_{j,\text{orig}}}$$

where

$$V = F_T = 80 \text{ N}$$

and

$$A_p = w_p t_p = 176 \text{ mm}^2$$

Substituting,

$$q_{\text{orig}} = \frac{(80 \text{ N})(10 \text{ GPa})(176 \text{ mm}^2)(19.5 \text{ mm})}{2191 \text{ N} \cdot \text{m}^2} \approx 1253 \text{ N/m}$$

Over the overlap length,

$$F_{\text{total,orig}} = 2F_{\text{plate,orig}} = 2q_{\text{orig}} L_b = 190.4 \text{ N}$$

The original design has 4 screws, so the shear force per screw is

$$V_{\text{screw,orig}} = \frac{F_{\text{total,orig}}}{4}$$

$$V_{\text{screw,orig}} = \frac{190.4 \text{ N}}{4} = \frac{47.6 \text{ N}}{\text{screw}}$$

The corresponding average shear stress is

$$\tau_{\text{screw,orig}} = \frac{V_{\text{screw,orig}}}{A_s}$$

Using

$$A_s = \frac{\pi(3 \text{ mm})^2}{4} = 7.07 \text{ mm}^2$$

gives

$$\tau_{\text{screw,orig}} = \frac{47.6 \text{ N}}{7.07 \text{ mm}^2} = 6.74 \text{ MPa}$$

Therefore, using the same shear-flow style estimate as the rest of Question 2,

$$V_{\text{screw,orig}} \approx \frac{47.6 \text{ N}}{\text{screw}}$$

and

$$\tau_{\text{screw,orig}} \approx 6.74 \text{ MPa}$$

This is still a small shear stress for a 3 mm metal screw. Therefore, even with the more conservative shear-transfer estimate, the screw shank itself is unlikely to be the controlling failure mode. The more likely concerns are bearing, local crushing, tear-out, delamination, or loosening around the screw holes.

3. Question 3

3.a. Given Values and Model

The leg is modeled as a fixed-free column. The spar end is fixed, while the other end is free to deflect and rotate. Therefore, for Euler buckling:

$$K = 2$$

The leg dimensions are:

$$L = 8 \text{ in} = 0.2032 \text{ m} \quad D_o = \frac{1}{2} \text{ in} = 0.0127 \text{ m}, \quad t = \frac{1}{16} \text{ in} = 0.0015875 \text{ m}$$

The inner diameter is:

$$D_i = D_o - 2t$$

The carbon fiber is $0^\circ/90^\circ$, so:

$$E = 40 \text{ GPa}$$

and

$$\sigma_u = 800 \text{ MPa}$$

3.b. Cross-Sectional Properties

The cross-sectional area of the hollow circular tube is:

$$A = \frac{\pi}{4}(D_o^2 - D_i^2) = 5.542 \times 10^{-5} \text{ m}^2$$

The area moment of inertia is:

$$I = \frac{\pi}{64}(D_o^4 - D_i^4) = 8.729 \times 10^{-10} \text{ m}^4$$

3.c. Direct Compressive Failure Check

If the leg failed by direct compressive stress, then:

$$\sigma = \frac{P}{A}$$

Setting $(\sigma = \sigma_u)$ gives:

$$P_{\text{material}} = \sigma_u A = 4.43 \times 10^4 \text{ N} = 44.3 \text{ kN}$$

Thus, direct compressive material failure would require a very large load.

3.d. Euler Buckling Check

For a column, the Euler critical load is:

$$P_{\text{cr}} = \frac{\pi^2 EI}{(KL)^2}$$

with ($K = 2$) for a fixed-free column,

Therefore:

$$P_{\text{cr}} = \frac{\pi^2 EI}{(2L)^2} \approx 2087 \text{ N}$$

In pounds-force,

$$P_{\text{cr}} \approx 469 \text{ lbf}$$

At this buckling load, the average compressive stress is

$$\sigma_{\text{cr}} = \frac{P_{\text{cr}}}{A} \approx 37.6 \text{ MPa}$$

This is much smaller than the ultimate strength of the carbon fiber:

$$37.6 \text{ MPa} \ll 800 \text{ MPa}$$

Therefore, the leg buckles before the material reaches its compressive ultimate strength.

3.e. Likely Failure Mode

The direct material failure load is

$$P_{\text{material}} = 44.3 \text{ kN}$$

The Euler buckling load is

$$P_{\text{cr}} = 2.09 \text{ kN}$$

Since

$$P_{\text{cr}} \ll P_{\text{material}}$$

the likely failure mode is

Euler buckling of the fixed-free carbon-fiber tube

Thus, the maximum axial load that one leg can take before failing is approximately

$$P_{\text{max}} \approx 2.09 \text{ kN}$$

3.f. Discussion

The loaded weight of the aircraft is

$$W = 89 \text{ N}$$

If the load were shared equally by four legs, then each leg would carry

$$W_{\text{leg}} = \frac{89 \text{ N}}{4 \text{ legs}} = \frac{22.25 \text{ N}}{\text{leg}}$$

The static factor of safety against Euler buckling would be

$$n = \frac{P_{\text{cr}}}{W_{\text{leg}}} = 93.8$$

This suggests that the legs are very safe under static axial compression. However, real landing loads can be much larger than the static aircraft weight because of impact loading, uneven load sharing, angled landings, side loads, and stress concentrations at the spar connection.

If the leg is loaded laterally as a cantilever rather than axially as a column, then bending stress would control. For a lateral tip load,

$$\sigma_{\text{max}} = \frac{PLc}{I}$$

where:

$$c = \frac{D_o}{2}$$

Solving for the lateral bending load,

$$P_{\text{bend}} = \frac{\sigma_u I}{Lc} = 541 \text{ N}$$

Therefore, under the fixed-free axial-column interpretation, the controlling load is

$$P_{\text{max}} \approx 2.09 \text{ kN}$$

with buckling as the likely failure mode. Under a lateral landing-load interpretation, bending could become more important and would give a lower approximate limit of

$$P_{\text{bend}} \approx 541 \text{ N}$$

4. Appendix Code

This notebook reproduces the calculations used in the Typst writeup for Questions 1, 2, and 3.

All calculations are performed in SI units unless otherwise noted.

```
In [35]: import math

def mm(x):
    return x * 1e-3

def inch(x):
    return x * 0.0254

def GPa(x):
    return x * 1e9

def MPa(x):
    return x * 1e6

def N_to_lbf(x):
    return x / 4.4482216152605

def m_to_mm(x):
    return x * 1000

def rad_to_deg(x):
    return x * 180 / math.pi

def print_result(name, value, unit="", sig=6):
    print(f"{name:<35s} = {value:.{sig}g} {unit}".rstrip())
```

```

In [36]: # Loads
F_T = 80.0 # N

# Geometry
t_p = mm(4.0) # plate thickness, m
w_p = mm(44.0) # plate width, m
t_w = mm(2.0) # spar wall thickness, m
d_o = mm(25.0) # spar outer side length, m
L_b = mm(76.0) # plate/spar overlap length, m
L_s = mm(400.0) # free spar length, m
L_total = L_b + L_s # total beam length, m

# Material properties
E_CF_0 = GPa(40.0) # Pa
E_CF_45 = GPa(10.0) # Pa
E_ASA = GPa(2.1) # Pa
sigma_u_0 = MPa(800.0) # Pa

print_result("F_T", F_T, "N")
print_result("t_p", t_p, "m")
print_result("w_p", w_p, "m")
print_result("t_w", t_w, "m")
print_result("d_o", d_o, "m")
print_result("L_b", L_b, "m")
print_result("L_s", L_s, "m")
print_result("L_total", L_total, "m")
print_result("E_CF_0", E_CF_0, "Pa")
print_result("sigma_u_0", sigma_u_0, "Pa")

F_T = 80 N
t_p = 0.004 m
w_p = 0.044 m
t_w = 0.002 m

```

d_o	= 0.025 m
L_b	= 0.076 m
L_s	= 0.4 m
L_total	= 0.476 m
E_CF_0	= 4e+10 Pa
sigma_u_0	= 8e+08 Pa

```
In [37]: # Spar section properties
d_i = d_o - 2*t_w
I_s = (d_o**4 - d_i**4) / 12

# Plate section properties
A_p = w_p * t_p
I_pc = w_p * t_p**3 / 12
d_p = (d_o + t_p) / 2

# Joint section property
I_j = I_s + 2*(I_pc + A_p*d_p**2)

# Flexural rigidities
EI_s = E_CF_0 * I_s
EI_j = E_CF_0 * I_j

print_result("d_i", d_i, "m")
print_result("I_s", I_s, "m^4")
print_result("I_s", I_s / 1e-12, "mm^4")
print_result("A_p", A_p, "m^2")
print_result("I_pc", I_pc, "m^4")
print_result("d_p", d_p, "m")
print_result("I_j", I_j, "m^4")
print_result("I_j", I_j / 1e-12, "mm^4")
print_result("E I_s", EI_s, "N*m^2")
```

```

print_result("E I_j", EI_j, "N*m^2")
print_result("I_j / I_s", I_j / I_s)

d_i = 0.021 m
I_s = 1.63453e-08 m^4
I_s = 16345.3 mm^4
A_p = 0.000176 m^2
I_pc = 2.34667e-10 m^4
d_p = 0.0145 m
I_j = 9.08227e-08 m^4
I_j = 90822.7 mm^4
E I_s = 653.813 N*m^2
E I_j = 3632.91 N*m^2
I_j / I_s = 5.55649

```

```

In [38]: theta_Lb = (F_T * L_b / EI_j) * (L_s + L_b/2)
y_Lb = (F_T * L_b**2 / EI_j) * (L_s/2 + L_b/3)

theta_s_free = F_T * L_s**2 / (2 * EI_s)
y_s_free = F_T * L_s**3 / (3 * EI_s)

theta_tip = theta_Lb + theta_s_free
y_tip = y_Lb + theta_Lb * L_s + y_s_free

print_result("theta(L_b)", theta_Lb, "rad")
print_result("theta(L_b)", rad_to_deg(theta_Lb), "deg")
print_result("y(L_b)", y_Lb, "m")
print_result("y(L_b)", m_to_mm(y_Lb), "mm")
print()
print_result("theta_s_free", theta_s_free, "rad")
print_result("y_s_free", y_s_free, "m")
print()
print_result("theta_tip", theta_tip, "rad")

```

```
print_result("theta_tip", rad_to_deg(theta_tip), "deg")
print_result("y_tip", y_tip, "m")
print_result("y_tip", m_to_mm(y_tip), "mm")
```

```
theta(L_b)          = 0.000733033 rad
theta(L_b)          = 0.0419997 deg
y(L_b)              = 2.86608e-05 m
y(L_b)              = 0.0286608 mm
```

```
theta_s_free        = 0.00978873 rad
y_s_free            = 0.00261033 m
```

```
theta_tip           = 0.0105218 rad
theta_tip           = 0.602852 deg
y_tip               = 0.0029322 m
y_tip               = 2.9322 mm
```

In [39]: # Equivalent expanded form for y_tip

```
y_tip_expanded = (
    F_T/EI_j * (L_s*L_b**2 + L_b**3/3 + L_s**2*L_b)
    + F_T*L_s**3/(3*EI_s)
)

print_result("y_tip from compact expression", y_tip, "m")
print_result("y_tip from expanded expression", y_tip_expanded, "m")
print_result("absolute difference", abs(y_tip - y_tip_expanded), "m")
assert abs(y_tip - y_tip_expanded) < 1e-15

y_tip from compact expression    = 0.0029322 m
y_tip from expanded expression   = 0.0029322 m
absolute difference               = 0 m
```

In [40]: # Back-of-the-book checks

```
y_lower = F_T * L_s**3 / (3 * EI_s)
```

```
theta_lower = F_T * L_s**2 / (2 * EI_s)

y_upper = F_T * L_total**3 / (3 * EI_s)
theta_upper = F_T * L_total**2 / (2 * EI_s)

print_result("y_lower", m_to_mm(y_lower), "mm")
print_result("theta_lower", theta_lower, "rad")
print_result("theta_lower", rad_to_deg(theta_lower), "deg")
print()
print_result("y_tip", m_to_mm(y_tip), "mm")
print_result("theta_tip", theta_tip, "rad")
print_result("theta_tip", rad_to_deg(theta_tip), "deg")
print()
print_result("y_upper", m_to_mm(y_upper), "mm")
print_result("theta_upper", theta_upper, "rad")
print_result("theta_upper", rad_to_deg(theta_upper), "deg")

assert y_lower < y_tip < y_upper
assert theta_lower < theta_tip < theta_upper

y_lower = 2.61033 mm
theta_lower = 0.00978873 rad
theta_lower = 0.560853 deg

y_tip = 2.9322 mm
theta_tip = 0.0105218 rad
theta_tip = 0.602852 deg

y_upper = 4.39882 mm
theta_upper = 0.0138618 rad
theta_upper = 0.794224 deg
```

```

In [41]: # Joint stress at fixed end
M_0 = F_T * L_total
c_j = d_o/2 + t_p
sigma_j = M_0 * c_j / I_j

# Free spar stress at start of free spar
M_Lb = F_T * L_s
c_s = d_o/2
sigma_s = M_Lb * c_s / I_s

sigma_max = max(sigma_j, sigma_s)
n_fos = sigma_u_0 / sigma_max

print_result("M(0)", M_0, "N*m")
print_result("c_j", c_j, "m")
print_result("sigma_j", sigma_j/1e6, "MPa")
print()
print_result("M(L_b)", M_Lb, "N*m")
print_result("c_s", c_s, "m")
print_result("sigma_s", sigma_s/1e6, "MPa")
print()
print_result("controlling sigma", sigma_max/1e6, "MPa")
print_result("factor of safety", n_fos)

M(0) = 38.08 N*m
c_j = 0.0165 m
sigma_j = 6.9181 MPa

M(L_b) = 32 N*m
c_s = 0.0125 m
sigma_s = 24.4718 MPa

```

controlling sigma = 24.4718 MPa
 factor of safety = 32.6907

```
In [42]: t_ASA = mm(5.0)

I_ASA = ((d_o + 2*t_ASA)**4 - d_o**4) / 12

# Rough original-joint stiffness:
# spar contribution + ASA collar contribution + two 45-degree plates
EI_j_orig = (
    E_CF_0 * I_s
    + E_ASA * I_ASA
    + 2 * E_CF_45 * (I_pc + A_p*d_p**2)
)

# Same beam formulas, but use EI_j_orig directly for the joint-region
stiffness.
theta_Lb_orig = (F_T * L_b / EI_j_orig) * (L_s + L_b/2)
y_Lb_orig = (F_T * L_b**2 / EI_j_orig) * (L_s/2 + L_b/3)

theta_tip_orig = theta_Lb_orig + theta_s_free
y_tip_orig = y_Lb_orig + theta_Lb_orig * L_s + y_s_free

print_result("t_ASA", t_ASA, "m")
print_result("I_ASA", I_ASA, "m^4")
print_result("EI_j_orig", EI_j_orig, "N*m^2")
print_result("EI_j_new", EI_j, "N*m^2")
print_result("EI_j_orig / EI_j_new", EI_j_orig / EI_j)
print()
print_result("theta_tip_orig", theta_tip_orig, "rad")
print_result("y_tip_orig", m_to_mm(y_tip_orig), "mm")
```

```

print_result("y_tip_new", m_to_mm(y_tip), "mm")
print_result("y_tip_orig / y_tip_new", y_tip_orig / y_tip)

t_ASA                      = 0.005 m
I_ASA                      = 9.25e-08 m^4
EI_j_orig                  = 1592.84 N*m^2
EI_j_new                   = 3632.91 N*m^2
EI_j_orig / EI_j_new       = 0.438447

theta_tip_orig             = 0.0114606 rad
y_tip_orig                 = 3.34445 mm
y_tip_new                  = 2.9322 mm
y_tip_orig / y_tip_new     = 1.14059

```

```

In [43]: # Simple equal-share screw shear

num_screws_new = 8
d_screw = mm(3.0)

V_screw_simple = F_T / num_screws_new
A_screw = math.pi * d_screw**2 / 4
tau_screw_simple = V_screw_simple / A_screw

print_result("V_screw_simple", V_screw_simple, "N/screw")
print_result("A_screw", A_screw, "m^2")
print_result("A_screw", A_screw/1e-6, "mm^2")
print_result("tau_screw_simple", tau_screw_simple/1e6, "MPa")

V_screw_simple          = 10 N/screw
A_screw                  = 7.06858e-06 m^2
A_screw                  = 7.06858 mm^2
tau_screw_simple         = 1.41471 MPa

```

```

In [44]: # Conservative shear-flow estimate for shear transfer between plates and spar

V = F_T

```

```

Q_one_plate = A_p * d_p
q = V * Q_one_plate / I_j          # N/m for one plate-to-spar
interface
F_plate = q * L_b
F_total_transfer = 2 * F_plate
V_screw_flow = F_total_transfer / num_screws_new
tau_screw_flow = V_screw_flow / A_screw

print_result("Q_one_plate", Q_one_plate, "m^3")
print_result("q", q, "N/m")
print_result("q", q/1000, "N/mm")
print_result("F_plate", F_plate, "N")
print_result("F_total_transfer", F_total_transfer, "N")
print_result("V_screw_flow", V_screw_flow, "N/screw")
print_result("tau_screw_flow", tau_screw_flow/1e6, "MPa")

Q_one_plate          = 2.552e-06 m^3
q                    = 2247.9 N/m
q                    = 2.2479 N/mm
F_plate              = 170.84 N
F_total_transfer     = 341.68 N
V_screw_flow         = 42.71 N/screw
tau_screw_flow       = 6.04224 MPa

```

In [45]: # Original socket design screw estimate using shear-flow style method

```

# Original design has 4 screws instead of 8
num_screws_orig = 4

d_p_orig = d_o / 2 + t_ASA + t_p / 2
q_orig = V * E_CF_45 * A_p * d_p_orig / EI_j_orig
F_plate_orig = q_orig * L_b
F_total_orig = 2 * F_plate_orig

```

```

V_screw_orig_flow = F_total_orig / num_screws_orig
tau_screw_orig_flow = V_screw_orig_flow / A_screw

print_result("d_p_orig", d_p_orig, "m")
print_result("d_p_orig", m_to_mm(d_p_orig), "mm")
print_result("EI_j_orig", EI_j_orig, "N*m^2")
print_result("q_orig", q_orig, "N/m")
print_result("q_orig", q_orig / 1000, "N/mm")
print_result("F_plate_orig", F_plate_orig, "N")
print_result("F_total_orig", F_total_orig, "N")
print_result("V_screw_orig_flow", V_screw_orig_flow, "N/screw")
print_result("tau_screw_orig_flow", tau_screw_orig_flow / 1e6, "MPa")

d_p_orig                = 0.0195 m
d_p_orig                = 19.5 mm
EI_j_orig               = 1592.84 N*m^2
q_orig                  = 1723.72 N/m
q_orig                  = 1.72372 N/mm
F_plate_orig            = 131.003 N
F_total_orig            = 262.005 N
V_screw_orig_flow       = 65.5013 N/screw
tau_screw_orig_flow     = 9.26653 MPa

```

```

In [46]: # Leg geometry
L_leg = inch(8.0)
D_o_leg = inch(0.5)
t_leg = inch(1.0/16.0)
D_i_leg = D_o_leg - 2*t_leg

# Tube properties
A_leg = math.pi/4 * (D_o_leg**2 - D_i_leg**2)
I_leg = math.pi/64 * (D_o_leg**4 - D_i_leg**4)

```

```
# Material
E_leg = E_CF_0
sigma_u_leg = sigma_u_0

print_result("L_leg", L_leg, "m")
print_result("D_o_leg", D_o_leg, "m")
print_result("t_leg", t_leg, "m")
print_result("D_i_leg", D_i_leg, "m")
print_result("A_leg", A_leg, "m^2")
print_result("I_leg", I_leg, "m^4")

L_leg = 0.2032 m
D_o_leg = 0.0127 m
t_leg = 0.0015875 m
D_i_leg = 0.009525 m
A_leg = 5.54211e-05 m^2
I_leg = 8.72937e-10 m^4
```

```
In [47]: P_material = sigma_u_leg * A_leg

print_result("P_material", P_material, "N")
print_result("P_material", P_material/1000, "kN")
print_result("P_material", N_to_lbf(P_material), "lbf")

P_material = 44336.9 N
P_material = 44.3369 kN
P_material = 9967.33 lbf
```

```
In [48]: K = 2.0

P_cr = math.pi**2 * E_leg * I_leg / ((K * L_leg)**2)
sigma_cr = P_cr / A_leg

print_result("P_cr", P_cr, "N")
print_result("P_cr", P_cr/1000, "kN")
```

```

print_result("P_cr", N_to_lbf(P_cr), "lbf")
print_result("sigma_cr", sigma_cr/1e6, "MPa")
print_result("sigma_u", sigma_u_leg/1e6, "MPa")
print_result("P_material / P_cr", P_material / P_cr)

```

```

P_cr                = 2086.58 N
P_cr                = 2.08658 kN
P_cr                = 469.082 lbf
sigma_cr            = 37.6496 MPa
sigma_u             = 800 MPa
P_material / P_cr   = 21.2486

```

In [49]:

```

W_aircraft = 89.0
num_legs = 4
W_leg = W_aircraft / num_legs
n_buckling_static = P_cr / W_leg

print_result("W_aircraft", W_aircraft, "N")
print_result("W_leg", W_leg, "N/leg")
print_result("static buckling FOS", n_buckling_static)

```

```

W_aircraft          = 89 N
W_leg               = 22.25 N/leg
static buckling FOS = 93.7789

```

In [50]:

```

c_leg = D_o_leg / 2
P_bend = sigma_u_leg * I_leg / (L_leg * c_leg)

print_result("c_leg", c_leg, "m")
print_result("P_bend", P_bend, "N")
print_result("P_bend", N_to_lbf(P_bend), "lbf")

```

```

c_leg                = 0.00635 m
P_bend               = 541.222 N
P_bend               = 121.672 lbf

```

4.a. Summary of Main Results

```
In [51]: summary = {
    "Q1 theta(L_b) [rad]": theta_Lb,
    "Q1 y(L_b) [mm]": m_to_mm(y_Lb),
    "Q1 theta_tip [rad]": theta_tip,
    "Q1 y_tip [mm]": m_to_mm(y_tip),
    "Q1 FOS bending": n_fos,
    "Q1 EC y_tip_orig [mm]": m_to_mm(y_tip_orig),
    "Q2 simple V_screw [N]": V_screw_simple,
    "Q2 simple tau_screw [MPa]": tau_screw_simple/1e6,
    "Q2 shear-flow V_screw [N]": V_screw_flow,
    "Q2 shear-flow tau_screw [MPa]": tau_screw_flow/1e6,
    "Q3 P_cr [kN]": P_cr/1000,
    "Q3 sigma_cr [MPa]": sigma_cr/1e6,
    "Q3 static buckling FOS": n_buckling_static,
    "Q3 lateral P_bend [N]": P_bend,
}
```

```
for key, value in summary.items():
    print(f"{key:<35s} {value:.6g}")
```

Q1 theta(L_b) [rad]	0.000733033
Q1 y(L_b) [mm]	0.0286608
Q1 theta_tip [rad]	0.0105218
Q1 y_tip [mm]	2.9322
Q1 FOS bending	32.6907
Q1 EC y_tip_orig [mm]	3.34445
Q2 simple V_screw [N]	10
Q2 simple tau_screw [MPa]	1.41471
Q2 shear-flow V_screw [N]	42.71
Q2 shear-flow tau_screw [MPa]	6.04224
Q3 P_cr [kN]	2.08658

Q3 sigma_cr [MPa]	37.6496
Q3 static buckling FOS	93.7789
Q3 lateral P_bend [N]	541.222